we can use sampling-based methods to *learn* the policy, known as reinforcement learning.

There is a large variety of RL approaches. First, we can separate value-function methods, which try to iteratively approximate the cumulative return specified in equation (1), and policy search, which tries to directly optimize some parametrized policy. Policy search shows promising results in real robotic applications (Kober and Peters, 2012). However, most work in RL utilizes value-function methods, on which we also focus in this survey.

Among value-function methods we should identify model-free versus model-based approaches. In model-free RL we iteratively approximate the value-function through temporal difference (TD) learning, thereby avoiding having to learn the transition function (which is usually challenging). Well-known algorithms are Q-learning (Watkins, 1989), SARSA (Rummery and Niranjan, 1994) and TD(λ) (Sutton, 1988). The update equation for Q-learning is given by:

$$Q(s,a) = Q(s,a) + \alpha\Big[r(s,a,s') + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big] \tag{4}$$

where α specifies a learning rate. With additional criteria for the learning and exploration parameters we can show this estimation procedure converges to the optimal value function (Sutton and Barto, 1998).

Model-based RL (Hester and Stone, 2012b) is a hybrid form of planning (like DP) and sampling (like TD learning). In model-based RL, we approximate the transition and reward function from the sampled experience. After acquiring knowledge of the environment, we can mix real sample experience with planning updates. We will write $M = \{\hat{T}, \hat{r}\}$ to denote the estimated model. Note that a model is derived from the full agent-environment interaction history at time-point t, as given by $g_t = \{s_0, a_0, s_1, a_1, s_2, ...s_{t-1}, a_{t-1}, s_t\}$.

A final aspect we have not yet discussed is the nature of the reward function. Traditional RL specifications assume an external reward signal (known as an 'external Critic'). However, as argued by Chentanez et al (2004), in animals the reward signal is by definition derived from neuronal activations, and the Critic therefore resides inside the organism. It therefore also incorporates information from the internal environment, making all reward 'internal'. Singh et al (2010) identifies two types of internal reward: extrinsic internal and intrinsic internal (we will omit 'internal' and simply use extrinsic and intrinsic from now on). Extrinsic reward is related to resources/stimuli/goals in the external world (e.g. food), possibly influenced by internal variables (e.g. sugar level). In RL terms, extrinsic reward explicitly depends on the content of the sensory information (i.e. the observed state). On the contrary, intrinsic reward is not dependent on external resources, but rather derived from the agent-environment history g and current model M. An example of intrinsic reward in animals is curiosity. Intrinsic reward is domain-independent, i.e. curiosity is not related to any external resource, but can happen at any state (dependent on the agent history g). In contrast, extrinsic reward for food will never occur in domains where food does not occur. Intrinsic motivation has been identified to serve a developmental role to organisms.

3 Survey structure and methodology

We intended to include all research papers in which reinforcement learning and emotion play a role. We conducted a systematic Google Scholar search for 'Emotion' AND 'Reinforcement Learning' AND 'Computational', and for 'Emotion' AND 'Markov Decision Process'. We scanned all abstracts for the joint occurrence of emotion and learning in the proposed work. When in doubt, we assessed the full article to determine inclusion. Moreover, we investigated all papers citing several core papers in the field, for example, Gadanho and Hallam (2001), Salichs and Malfaz (2012), Broekens et al (2007a) and Marinier and Laird (2008). This resulted in 52 papers included in this survey. A systematic overview of these papers can be found in Table 9 and 10.

The proposed taxonomy of emotion elicitation, type and function is shown in Table 1, also stating the associated subsection where each category is discussed. The elicitation and function categories are also visually illustrated in Figure 1, a figure that is based on the motivated RL illustration (with internal Critic) introduced in Chentanez et al (2004). Figure 1 may be useful to refer back to during reading to integrate the different ideas. Finally, for each individual paper the reader can verify the associated category of emotion elicitation, type and function through the colour coding in the overview Table 9.

There is one important assumption throughout this work, which we want to emphasize here. We already introduced the distinction between extrinsic and intrinsic motivation in RL at the end of the last section. Throughout this work, we parallel extrinsic motivation with homeostasis (Section 4.1), and intrinsic motivation with appraisal (Section 4.2). The extrinsic/intrinsic distinction is clearly part of the RL literature, while homeostasis and especially appraisal belong to the affective modelling literature. We group these together, as the concept of extrinsic motivation is frequently studied in combination with homeostasis, while intrinsic motivation shows large overlap with appraisal theory. We will identify this overlap in the particular sections. However, the point we want to stress is that the concepts are not synonyms. For example, it is not clear whether some intrinsic motivation or appraisal dimensions also show homeostatic dynamics (a point at which we tend to disagree with Singh et al (2010)). However, a full discussion of the overlap and difference moves towards psychology, and is beyond the scope of our computational overview. We merely identify the overlap we observed in computational implementations, and therefore discuss both extrinsic/homeostasis and intrinsic/appraisal as single sections.

Table 1 Overview of categories in emotion elicitation, emotion type and emotion function. The number before each category identifies the paragraph where the topic is discussed. Emotion elicitation and function are also visually illustrated in Figure 1.

Emotion elicitation	Emotion type	Emotion function
4.1 Homeostasis and extrinsic motivation	5.1 Categorical	6.1 Reward modification
4.2 Appraisal and intrinsic motivation	5.2 Dimensional	6.2 State modification
4.3 Value/reward-based		6.3 Meta-learning
4.4 Hard-wired		6.4 Action selection
		6.5 Epiphenomenon

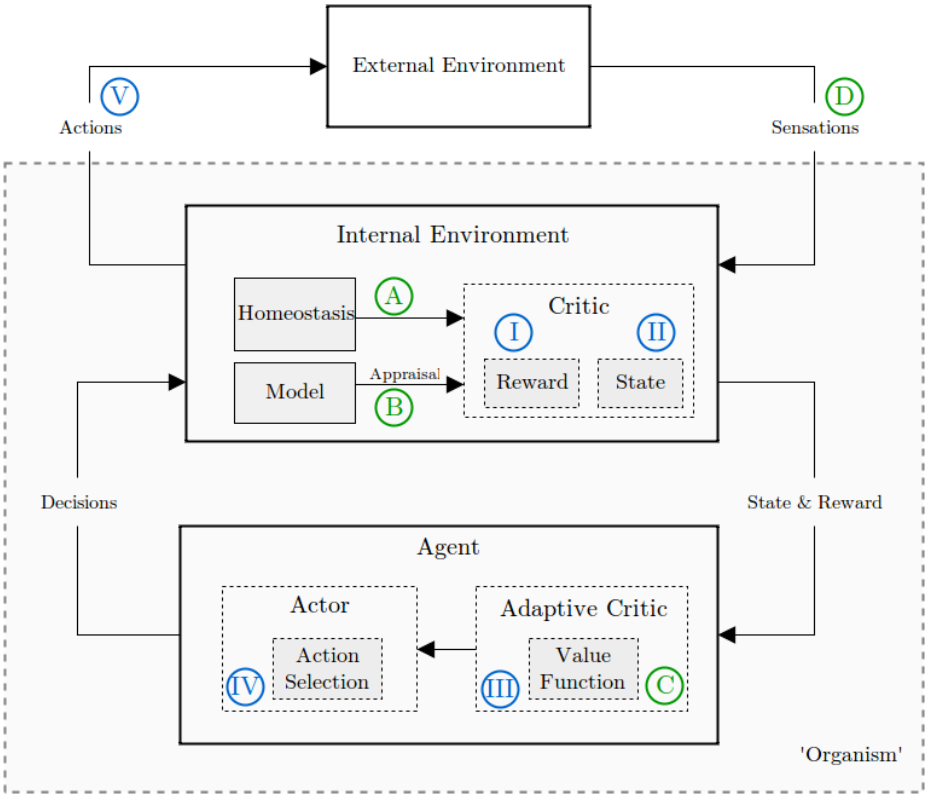

Fig. 1 Schematic representation of motivated reinforcement learning based on Chentanez et al (2004). Although traditional RL assumes an external Critic (to provide the reward signal), this actually happens inside the brain of real-world organisms. Thereby the Critic also incorporates, apart from external sensations, internal *motivations* to determine the current reward and state. Motivations have been derived from homeostatic variables and/or internal models. The Critic then feeds the state and reward to the Agent. The Agent usually learns a value function (Adaptive Critic) and determines the next action (Actor). Note that ordinary RL, in which the reward is a fully external stimulus, is still a specific case of this scheme (with the Critic as identity function). Emotion elicitation (green) has been associated to A) Homeostasis and extrinsic motivation (paragraph 4.1), B) Appraisal and intrinsic motivation (4.2), C) Reward and value function (4.3) and D) Hard-wired connections from sensations (4.4). Subsequently, the elicited emotion may also influence the learning loop. Emotion function (blue) has been linked to I) Reward modification (6.1), II) State modification (6.2), III) Meta-learning (6.3), IV) Action selection (6.4) and finally as V) Epiphenomenon (6.5).

4 Emotion elicitation

We identify four major categories of emotion elicitation: extrinsic/homeostatic (4.1), intrinsic/appraisal (4.2), value function and reward-based (4.3), and finally hard-wired (4.4).